PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AI-based drop-out prediction and counseling system

A PROJECT REPORT

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BACHELOR OF TECHNOLOGY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "AI-based drop-out prediction and counseling system" is a bonafide work of "KADAVAKOLLA LIKHITHA (20221IST0021), S ABHINAV (20221IST0064), GANGULA SURESH REDDY (20221IST0067)", who have successfully carried out the project work and submitted the report for partial fulfillment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE and Technology during 2025-26.

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We the students of final year B.Tech in INFORMATION SCIENCE and TECHNOLOGY at Presidency University, Bengaluru, named KADAVAKOLLA LIKHITHA, S ABHINAV, GANGULA SURESH REDDY, hereby declare that the project work titled "AI-BASED DROP-OUT PREDICTION AND COUSELLING SYSTEM" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in Information Science and Technology during the academic year of 2025-26.

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ABSTRACT

Student dropout is a major issue faced by educational institutions worldwide, affecting academic outcomes and institutional reputation. Factors such as poor academic performance, lack of interest, financial constraints, and personal issues contribute to this problem. Early identification of at-risk students is essential to enable timely interventions and improve retention rates. However, traditional methods of identifying such students are often manual, time-consuming, and less accurate, making it difficult for institutions to respond effectively.

The proposed **AI-Based Student Dropout Prediction System** aims to develop an intelligent and automated model to predict the likelihood of student dropout using machine learning techniques. The system analyzes historical student data, including academic performance, attendance, demographics, and other relevant factors, to identify patterns associated with dropout behavior. By applying classification algorithms, it can accurately detect at-risk students and provide early alerts to educators and administrators.

The system is developed using Python along with various machine learning libraries and involves processes such as data collection, preprocessing, feature selection, model training, and evaluation. It includes a user-friendly interface that allows users to input student data and obtain predictions easily, making it accessible even to non-technical users.

A key feature of the system is its ability to provide data-driven insights into the factors influencing student dropout. This enables institutions to take appropriate measures such as academic support, counseling, or financial assistance. The system can also be integrated with existing learning platforms for continuous monitoring of student performance. It is scalable, efficient, and adaptable across different educational environments, including schools, colleges, and universities.

By leveraging artificial intelligence and predictive analytics, this system promotes a proactive approach to education management. It ultimately aims to improve student retention and academic success through early identification and timely intervention

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List of Abbreviations

Abbreviation	**Full Form**
SDGs	Sustainable Development Goals
PWA	Progressive Web Applications
UI/UX	User Interface and User Experience
API	Application Programming Interface
ORM	Object Relation Mapper
UAT	User Acceptance testing
SQL	Structured Query Language
HDL	Hardware Description Layer
HTTPS	HyperText Transfer protocol Secure
SSL	Secure Socket Layer

Chapter 1

INTRODUCTON

1.1 Background

Education plays a great role in personal and social development. But, one of the problems that most learning institutions encounter is student dropout. Academic problems or financial constraints, or interests, can make students quit their studies. As how the Artificial Intelligence (AI) and the Machine Learning (ML) are developed, student data may be examined and the trends associated with dropping out can be determined. These technologies are used in the proposed system to predict at-risk students so they can be intervened with early.

1.2 Statistics of Project

High levels of dropout of students in most institutions are as a result of different factors. It has been observed that:

- Lowly academic performance increases the chances of dropping out among students.
- The poor attendance is one of the significant indicators of risk of dropping out.
- Discontinuation is also caused by financial and personal reasons.

This project will deal with these problems by examining the student data to predict a possibility of dropping out at an early stage..

1.3 Prior Existing Technologies

Already developed systems such as Learning Management Systems (LMS) assist in monitoring the student performance and managing courses. There are also some platforms that offer basic data analysis.

These systems have limitations however:

- Pay more attention to monitoring and less to prediction.
- Lack developed machine learning capabilities.
- Demand human examination to make a decision.

The given system addresses these shortcomings with the help of AI-based predictive models.

1.4 Proposed Approach

The suggested system employs machine learning to forecast the dropout of students. Its functionality is to analyze the data on the students like academic performance, attendance and so forth.

Key features include:

- Preprocessing of the data and training of the model.
- At-risk student prediction.
- An interface friendly to users to input and output.

Applications:

- Applicable in schools and university to spy on students.
- Assistance in early preventive measures.

Limitations:

- Depends on data quality Needs adequate information to make precise forecasts.

1.5 Objectives

The primary aims of the project are:

- To create a smart system to predict student dropout on machine learning methods.
- To examine different variables which include academic performance, attendance and personal background that affect the retention of students.
- To determine the students who are in danger of early dropout.
- In order to create an easy to use interface enabling easy interaction and prediction.
- To present evidence-based information that can assist teachers and managers to make relevant decisions.
- To minimize levels of dropout and enhance general performance and performance of students.

1.6Sustainable Development Goals (SDGs)

The following goals of the United Nations Sustainable Development Goals (SDGs) are in line with this project:

- **SDG 4: Quality Education** - The system assists in enhancing the quality of education

by establishing at-risk students and facilitating timely intervention to make sure that the learning process is inclusive and equitable.

- **SDG 10: Reduced Inequalities**- When analyzing a number of socio-economic and academic factors, the system assists with identifying the students with disadvantaged background and helps them with the required interventions, facilitating equal opportunities.
- **SDG 9: Industry, Innovation, and Infrastructure** - SDG 9 is the project which involves the latest technologies such as Artificial Intelligence and Machine Learning which improves the educational systems, which enhances innovation and digital transformation in the education sector.

Chapter 2

LITERATURE REVIEW

2.1 Dekker et al - Utilized to forecast student dropout

Dekker et al came up with a data mining method to gauge the likelihood of student dropout based on academic performance or data. They employed techniques of classification to determine risky students in their model. The research demonstrated that machine learning is applicable to the successful prediction of dropout with and reasonable accuracy. Nevertheless, the model primarily concentrated on academic information and did not take into account the other aspects such as the individual or socio-economic status. This study emphasizes the significance of predictive models in the field of education and justifies the use of machine learning in this study.

2.2 Kotsiantis et al. - Techniques of machine learning in education.

Kotsiantis and colleagues also tested different machine learning algorithms including Decision Trees, Neural Networks and Naive Bayes to predict student performance. They conducted a comparison between the various models made the conclusion that not one of the algorithms works best in every instance. The weakness of this work, however, is that it pays more attention to performance prediction, as opposed to dropout prediction. The study aids in the finding of appropriate algorithms in the construction of a quality prediction system. Rumberger - Factors Influencing Student dropout

2.3 Rumberger – Factors Influencing Student Dropout

Rumberger researched the key elements leading to the student dropout such as the academic performance, family background, and institutional support. The study stressed that a dropout is a complex phenomenon that cannot be predicted with the help of one parameter. The paper is however purely theoretical and lacks a technical application. This justifies the use of a system that takes into consideration a variety of features as used in this project.

2.4 Herzog - Early Warning Systems to Student Retention

Herzog came up with an early warning system to monitor students who are likely to drop out. The system utilized the data of students to create alerts to teachers. The outcome of these was that student retention was improved when there were early interventions. Nonetheless, the system was not automated and needed to be monitored manually. This is the reason behind the creation of an automated system using AI in the proposed project.

2.5 Gray et al. - Predictive Analytics in Education.

Gray et al discussed the predictive analytics used to enhance the success of students. Their research showed that the use of data-driven models can assist institutions in making improved decisions. The system was good as it gave valuable insights, but it was also big data-intensive and needed complicated infrastructure. This underscores the necessity of having a simpler and scalable solution such as the one suggested in this project

Summary of Literatures reviewed

Table 2.1 Summary of Literature Review

Author(s)	**Title/focus**	**Approach/methods**	**key findings**	**limitations**	**relevance to our project**
Dekker et al.	Dropout prediction using data mining	Classification models	Accurate prediction of dropout	Focused only on academic data	Supports ML-based prediction
Kotsiantis et al.	ML in education	Compared multiple algorithms	Different models give varying results	Not focused on dropout	Helps in model selection

Rumberger	Droupout factors	Theoretical analysis	Multiple factors influence dropout	No implementation	Supports multi-factor analysis
Gray et al.	Predictive analytics	Data-driven models	Improves decision-making	Requires large data	Supports analytics use
Herzog	Early warning system	Data-based alerts	Improves retention	Manual process	Motivates automation

Chapter 3
METHADOLOGY

Introduction

This chapter states how the AI-Based Student Dropout Prediction System was developed. The system is systematic and uses data of the students to compute predictive and precise results. Such steps are part of the development process in which the gathered data is collected, preprocessed, features are selected, the model is trained, evaluated, and predicted. At the start, student-based data, academic data, attendance data, and demographic data, are obtained using the accessible datasets. These data are then cleaned and processed to eliminate the inconsistencies and deal with the missing data. Once preprocessing is done, significant characteristics that affect the dropout of students will be chosen in order to enhance the performance of the model. The model is trained on past data with the help of machine learning algorithms to learn various patterns and connections among varying attributes. Accuracy and precision are then used to test the trained model to determine reliability. After the model is developed, it is incorporated into the system in which the user may enter the data about students and receive predictions. System architecture provides the system to communicate with the frontend, the backend, and the prediction model naturally. Generally, the methodology is efficient, scalable and accurate in determining at-risk students and has assisted institutions to make timely decisions in minimizing the drop out rates.

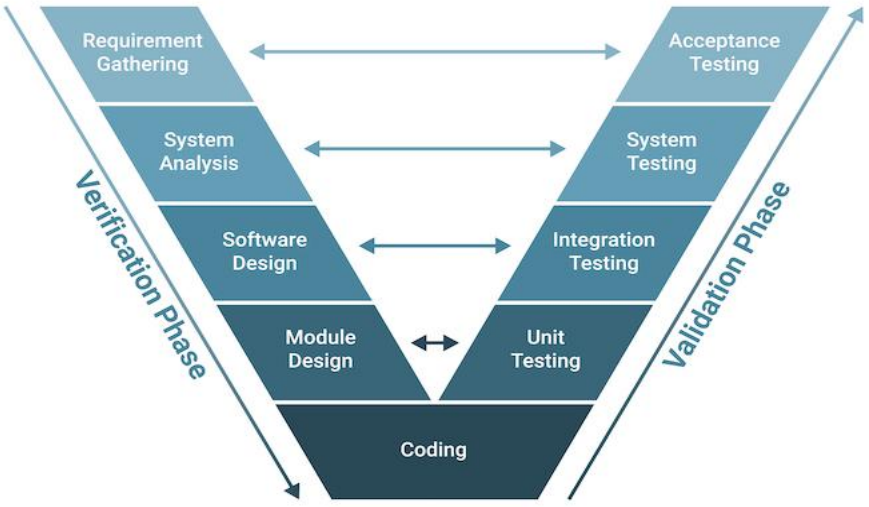


Figure 3.1 V-Model Methodology

- V-Model is a stipulated software development life cycle that demonstrates the correlation between development and testing processes.
- The left side of the V contains such stages as the requirement analysis, system design and implementation.
- Testing stages involved in the right side of the V are unit testing, integration testing and system testing.
- Every development phase is linked with a testing phase whereby it has to be checked that it is correct.
- This model assists to identify the mistakes at the initial stage and enhances the overall system quality.

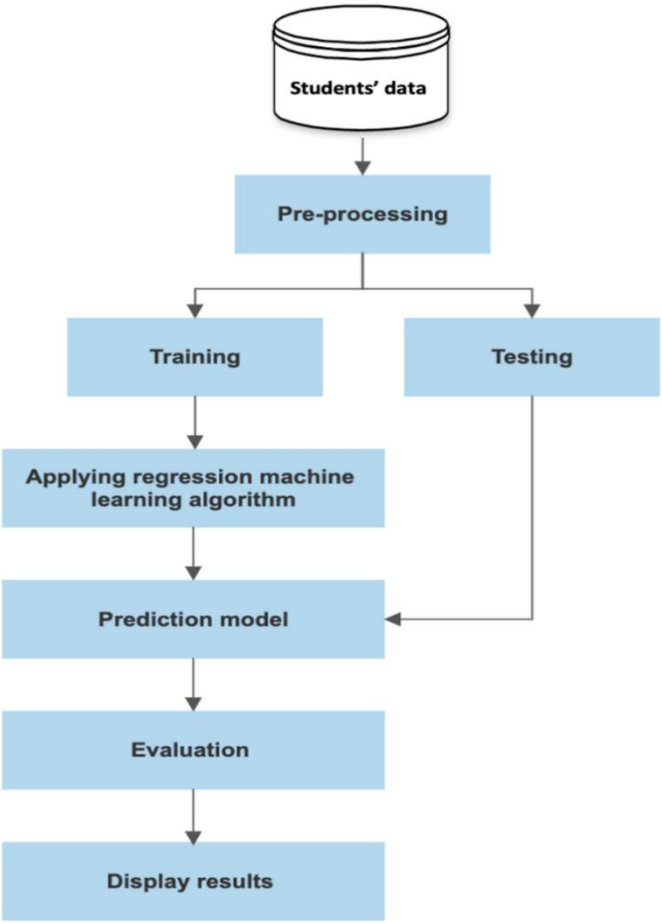


figure 3.2 Project Flow Diagram

- This can be initiated by the user, who supplies student information into the interface.
- The input is relayed to the backend system through the frontend.

- The backend data preprocesses and transfers it to the trained machine learning model.
- The data is analyzed using the model and it produces a prediction result.
- The outcome is presented to the user and the process of the system is completed.

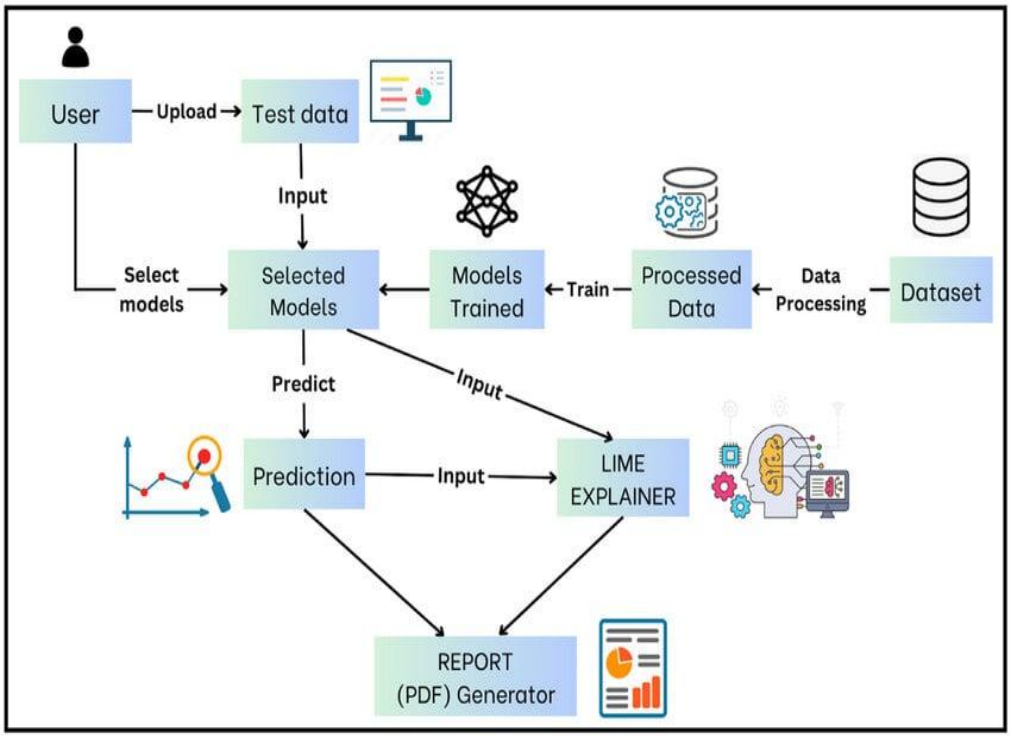


figure 3.3 Backend workflow diagram

- The backend begins by providing user authentication to provide protection to access to the system.
- The validation and checking of input data against correctness are carried out before the processing.
- The system uses preprocessing tasks and forwards information to the prediction model.
- The model then works with the input and comes up with a prediction result.
- The backend provides the answer to the frontend and provides a smooth

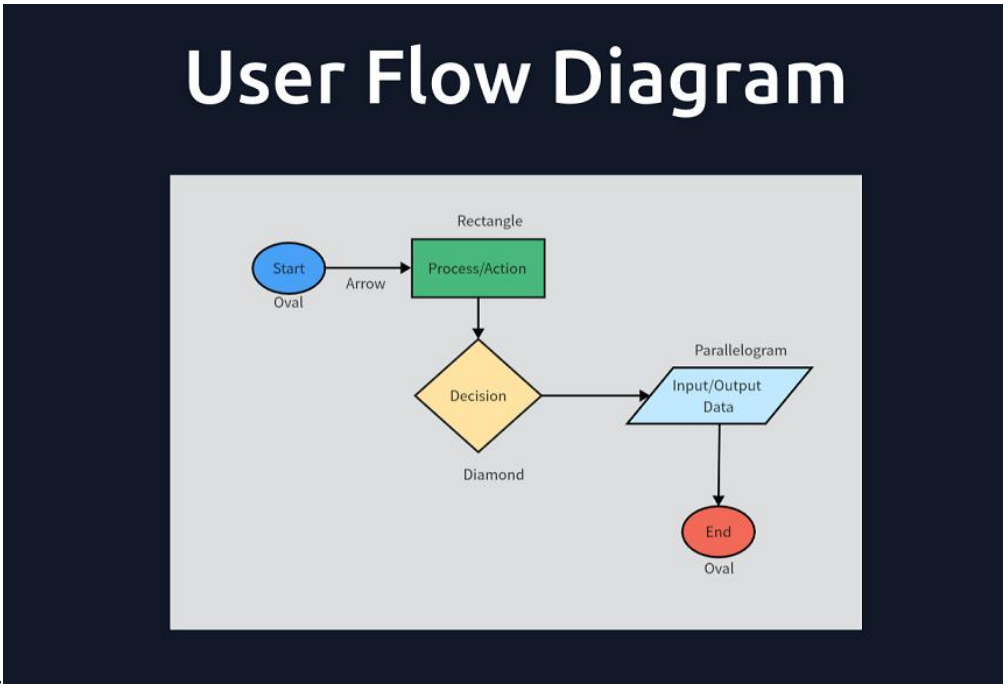
flow . 

figure 3.4 Frontend User interaction Flow

- This begins when the user enters the system via the interface.
- The user enters the device with the help of credentials.
- Upon logging in the dashboard will be revealed and choices will be available.
- The user will be able to input or remove student information. The system presents the results of prediction depending on an input.

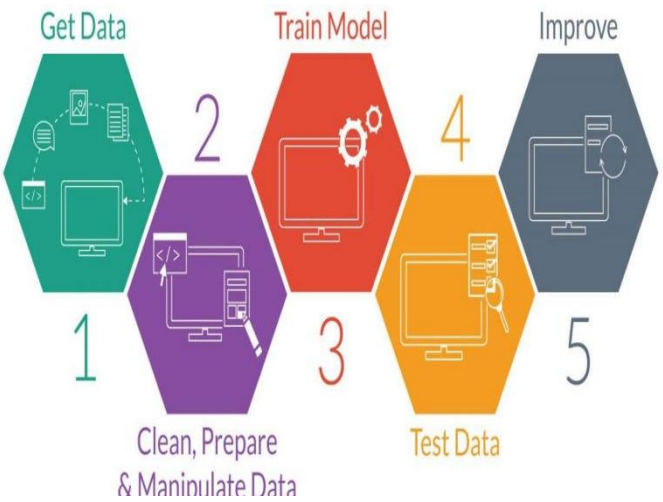


Figure 3.5 – Backend Architecture

- The frontend relays the request to the backend system.
- All processing and communication is done by the backend/API.
- Input is ready to be predicted through data processing.
- There are results of the machine learning model.
- Student data is stored and retrieved in the database.

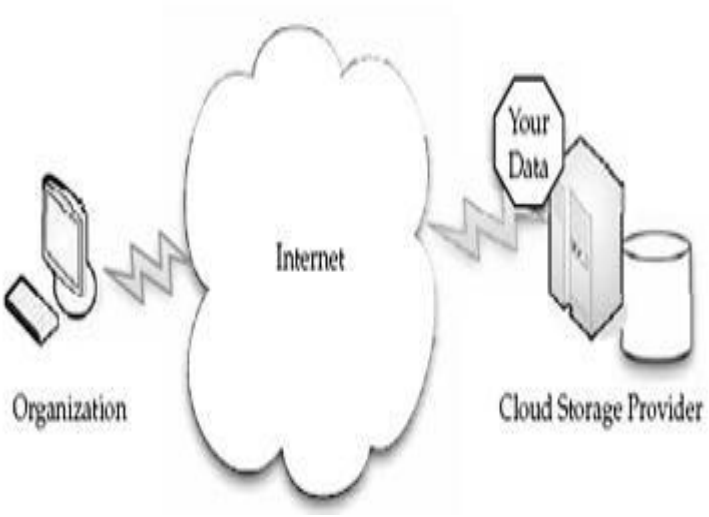


Figure 3.6 – Data Synchronization Workflow

- The data is initially stored within the system.
- Changes in data are monitored by the sync manager.
- Synchronization begins when there is internet.
- Information is transmitted to the server via API.
- The cloud database is updated with the newest information.

Chapter 4
Project Management

4.1 Introduction

Makes sure that all the stages of the project such as data preprocessing, The chapter explains how to plan and manage the AI-Based Student Dropout Prediction System. It incorporates the main points like project scheduling, risk, environmental, and resource allocation that are crucial to the successful project completion. Well planned project management model development, system design, testing, and implementation are conducted in a systematic and well structured manner within the allocated timeframe.

There is a clear timetable to monitor the development of every step and meet the deadline of the job. Risk analysis is conducted to determine possible challenges including data quality problems, limitations in the accuracy of the model, and technical challenges, as well as the relevant mitigation measures. PESTLE enables the study of the environment to determine external forces such as technology, legal restrictions, and social acceptance that could affect the project.

Also, efficient resource allocation is important in the management of human, software, hardware requirements and time. Proper allocation of such resources makes the project run smoothly and without delays and inefficiencies. Comprehensively, this chapter explains the need of organized planning and administration to provide an effective, predictable, and efficient system of student dropout prediction.

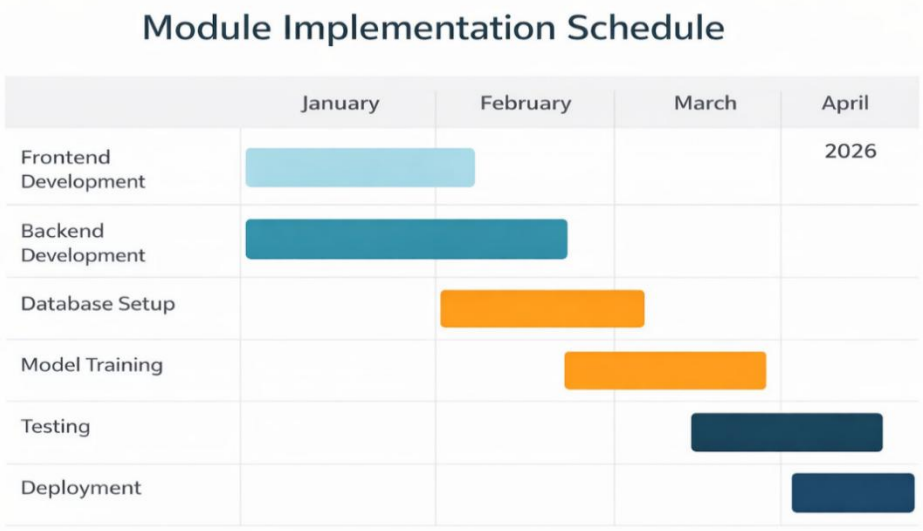


Figure 4.1: Module Implementation Schedule

The project was worked out in a systematic way by breaking down the work into various stages. Every stage was allotted a certain amount of time so that the flow would be even.

- **Frontend Development**: To be done in January and February to design the user interface.
- **Backend Development**: This was done in the months of February and March to process and integrate the models.
- **Database Setup**: This was done in March to store student data and results.
- **Model Training**: March and April machine learning algorithms.
- **Testing**: This was done in April to test the performance of the system.
- **Deployment**: Last stage in which the system is prepared to use it.

This schedule is useful in monitoring the progress and completion of the project on time

Figure 4.2 PESTLE Analysis

PESTLE analysis is used to understand external factors that may influence the project.

- **Political:** Government policies related to education and digital learning platforms.
- **Economic:** Cost of implementation and availability of resources.
- **Social:** Acceptance of AI-based systems by students and institutions.
- **Technological:** Availability of machine learning tools and technologies.
- **Legal:** Data privacy, security, and regulations related to student data.
- **Environmental:** Promotion of digital systems reduces paper usage and supports sustainability.

This analysis helps in identifying external factors affecting the project.

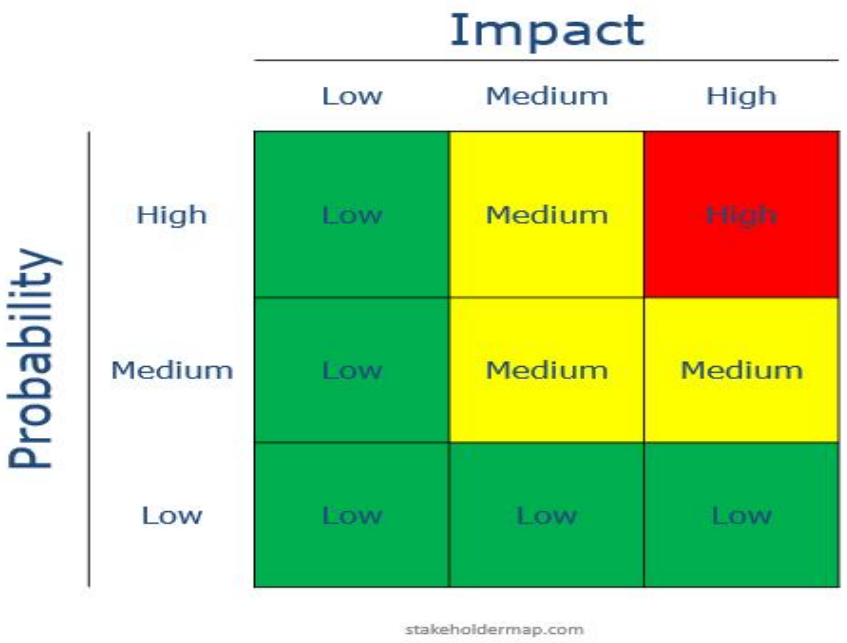


Figure 4.3 Risk Matrix

Risk identification and classification Risk is assessed and categorized in the risk matrix according to the likelihood and effect.

- **Low Risk**: Problems with low effect and low likelihood.
- **Medium Risk**: Medium impact which can bring changes to a project.
- High Risk: These are critical risks that may have a serious impact on the project.

Examples of risks:

- Data quality issues Model accuracy limitations
- Technical implementation challenges
- Time constraints Risk matrix would be used to prioritize risks and preventive measures.and coordinating the work of the team.

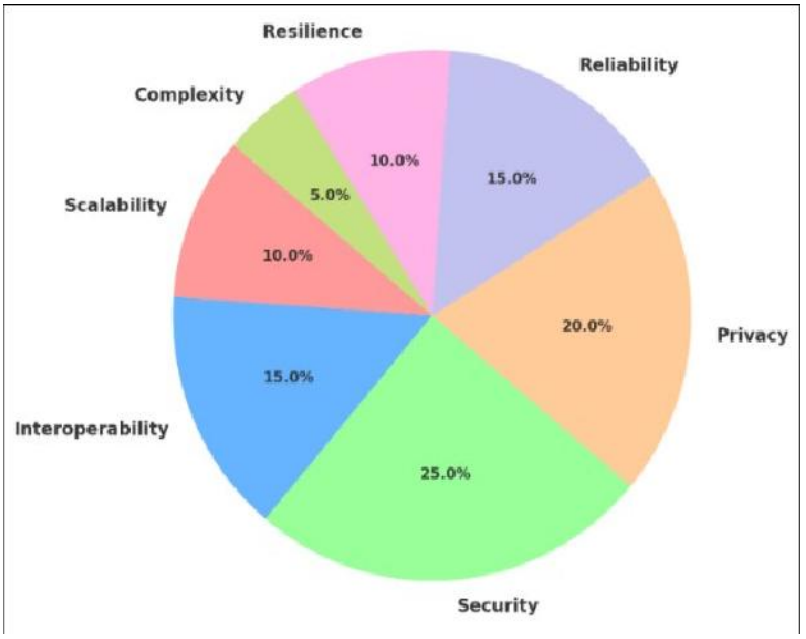


Figure 4.4 Resource Allocation Chart

Resource allocation is to provide an efficient use of available resources in the project.

- **Human Resources**: Development, testing, and documentation team members.
- **Software Resources**: Python, machine learning libraries, development tools.
- **Hardware Resources**: Computer systems and internet connection.
- **Time Resources**: Each stage should be scheduled properly to prevent time loss.

Chapter 5

Analysis and Design

5.1 Requirements

The requirements of the AI-Based Student Dropout Prediction System are established to illustrate the purpose of the system, its functionality, and limitations. The table below provides an overview of the major requirements of the system, both software and hardware.Summarizing requirements.

Table 5.1 Summarizing Requirements

Requirement Type	Description
Purpose	The system aims to predict student dropout using machine learning techniques to help institutions take early preventive measures.
Behaviour	The system collects student data, processes it, and predicts whether a student is at risk of dropping out.
System Management	The system requires minimal manual intervention. Users can input data and receive predictions automatically.
Data Analysis	The system performs data preprocessing and analysis to identify patterns affecting student dropout.
Application Deployment	The application can be deployed as a web-based system accessible through browsers.
Security	The system ensures secure login, data privacy, and protection of student information.
System HW Requirement	The system should run on basic hardware such as laptops or desktops with internet access.
System SW Requirement	The system is developed using Python with libraries like Pandas, NumPy, and Scikit-learn.
User Interface	The interface should be simple, user-friendly, and easy to navigate for all users.

5.2 Block Diagram

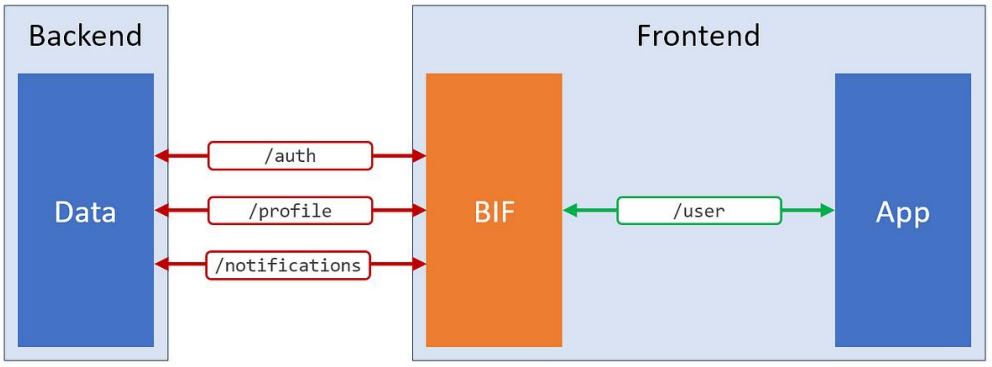


Fig 5.1 Functional block diagram

- The user interface enables users to interact with the system through inputting student information and looking at prediction outcomes. It is made to be user friendly and easy to use.
- The backend is the central processing unit that processes the requests, regulates the flow of data, and links various elements of the system effectively.
- The machine learning model takes the input data and runs patterns to determine whether a student is at risk of dropping out or not.
- The database receives the information about students, the processed data and the prediction outcomes; the data could be used and analyzed in the future.
- Every element is linked to another, which allows communication and productive work of the system.

System Flowchart

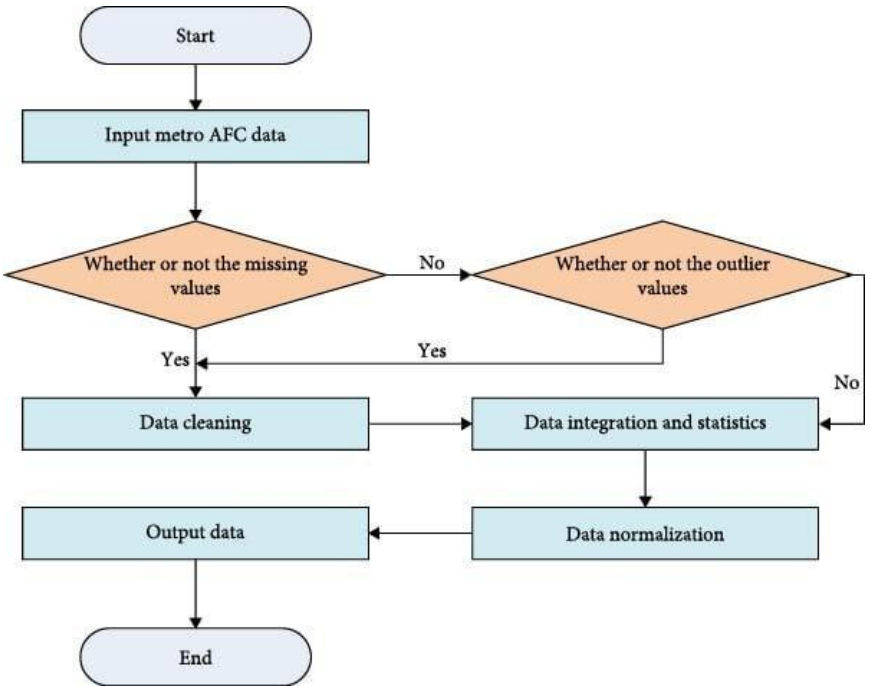


figure 5.2 System Flowchart for Data Synchronization

- This process starts with the entry of student data into the system by the user. This information consists of academic and personal information needed in prediction.
- Preprocessing stage involves cleaning and transformation of the data by taking care of the missing values, converting the data to a format that is useful.
- The processed data is analyzed with the help of the machine learning model and patterns of student dropout are identified. According to the analysis, the system will produce a prediction that will show whether the student will be at risk or not.
- Lastly, the output is shown to the user and this concludes the system workflow.All in all, the flowchart is a very effective and fidel process that guarantees a smooth coordination of the users-system components.

5.3 Use Case Diagram

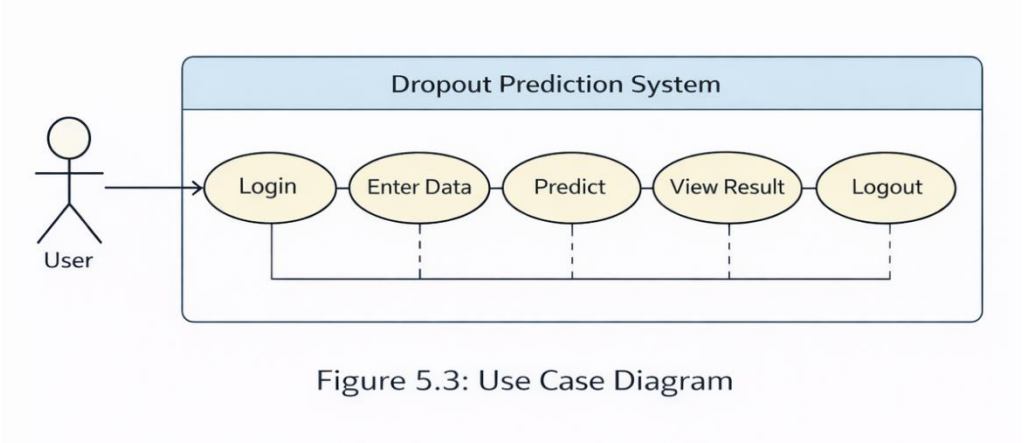
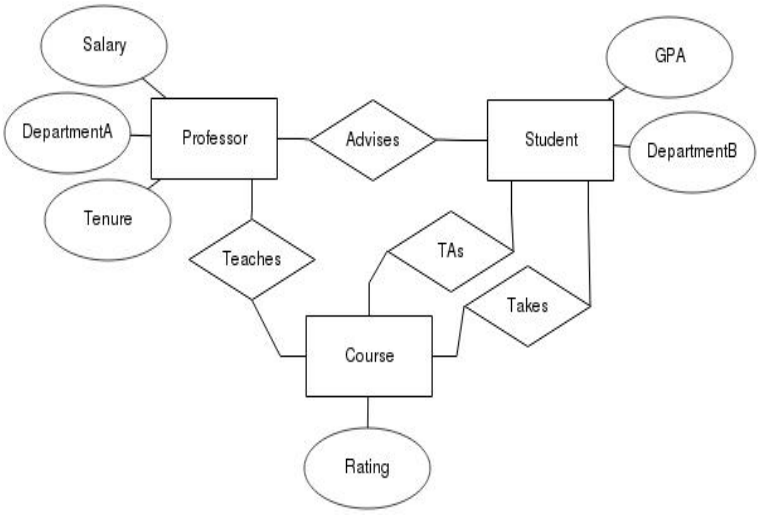


figure 5.3 Use Case Diagram

- The user is the main actor who communicates with the system to achieve different tasks like to log in and input data.
- The login use case has a guarantee that only authorized people can access the system and therefore, security and privacy is preserved.
- Enter data use case enables users to enter student data that is needed to predict.
- The view result use case allows the users to view the result of prediction made by the system.
- This illustration shows how users will treat various functionalities of the system.



5.4 Entity based Relationship Diagram

- The sequence diagram shows the interaction between different components in a step-by-step manner. It represents how requests and responses flow in the system.
- When the user logs in, he/she submits his/her credentials to the backend that authenticates them with the database and responds with a successful response.
- The system transmits new data to the server in a synchronization module in data synchronization.
- The request is processed by the server and the database is updated, which guarantees the consistency of data.
- The diagram assists in comprehending what order of operation and communication between parts takes place.

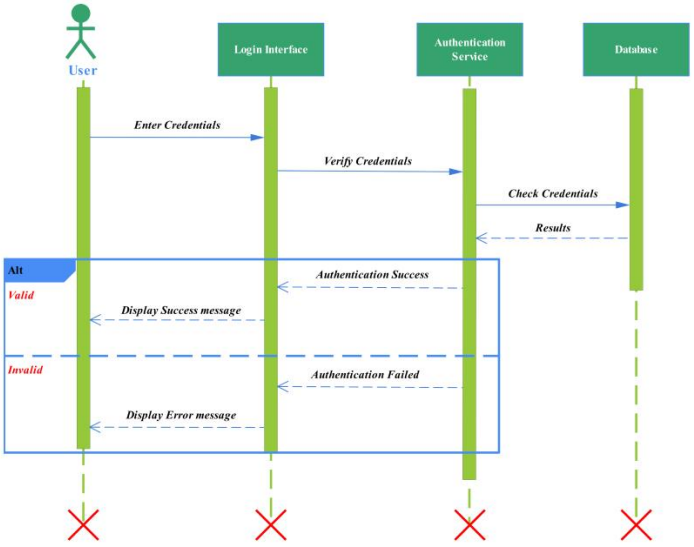


figure 5.5 Sequence Diagram (Login & Data Sync)

- The activity diagram is a model that shows the sequence of steps undertaken by the user in the system. It demonstrates the way the process works out until completion.
- This is initiated by user login, thus preventing unauthorized persons to gain access to the system.
- Once the user has logged on, they interact with the system through entering student data and making predictions.

- The system takes the request and produces results according to the machine learning model.
- This process concludes upon presenting the result, and the flow of the activity comes to an end.

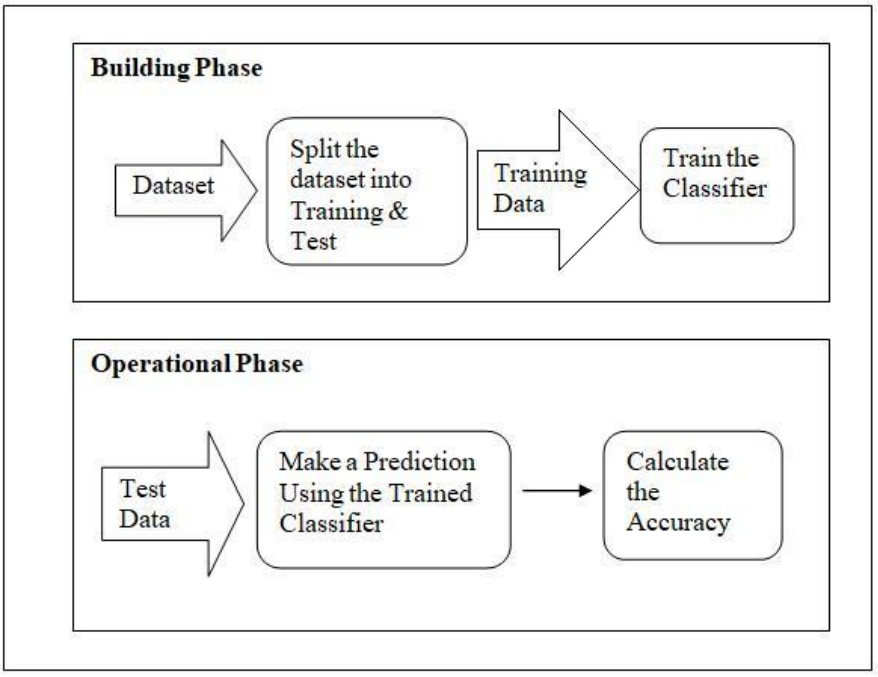
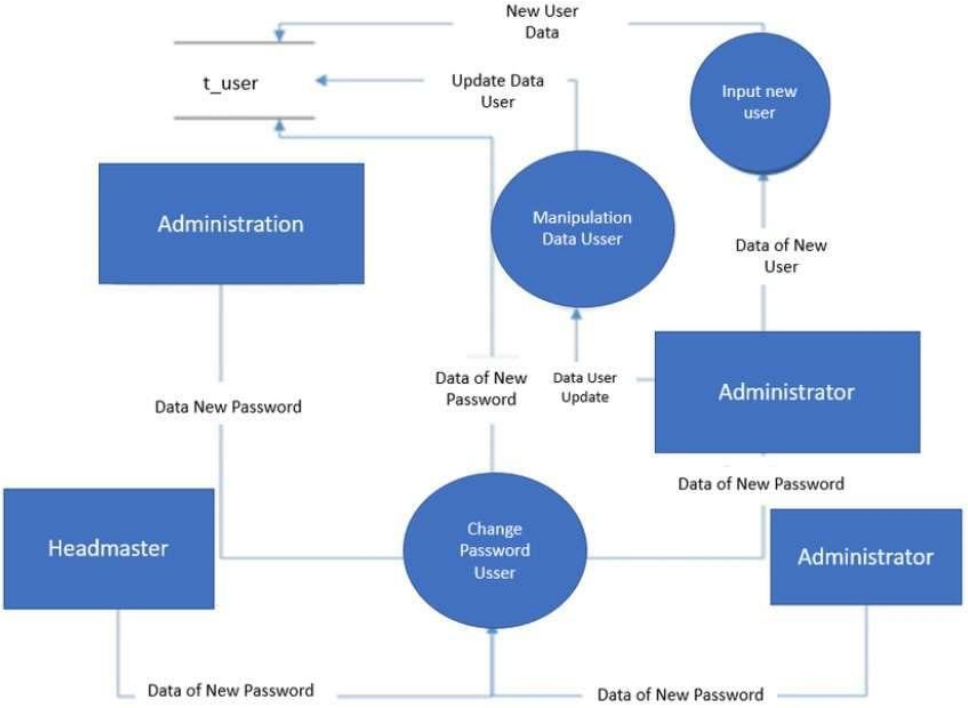


Figure 5.6 Activity Diagram

- The activity diagram represents the flow of actions performed by the user within the system. It shows how the process progresses from start to end.
- The process begins with user login, which ensures that only authorized users can access the system.
- After login, the user interacts with the system by entering student data and requesting predictions.
- The system processes the request and generates results based on the machine learning model.
- The process ends after displaying the result, completing the activity flow.



5.7 Data Flow Diagram (Level 1)

- The user provides input data such as student details required for prediction. This acts as the starting point of the system.
- The input data is processed and cleaned in the preprocessing stage to ensure accuracy and consistency.
- The machine learning model analyzes the processed data and identifies patterns related to dropout.
- Based on the analysis, the system generates a prediction result indicating the risk level.
- The final output is displayed to the user in an understandable format.

Chapter 6

Hardware, Software and Simulation

6.1 Hardware

The following chapter provides an overview of the hardware and software specifications of the AI-Based Student Dropout Prediction System development and testing. As the system is a web-based application, it does not require specialized hardware components. The emphasis lies on compatibility, scalability, and usability across various devices.

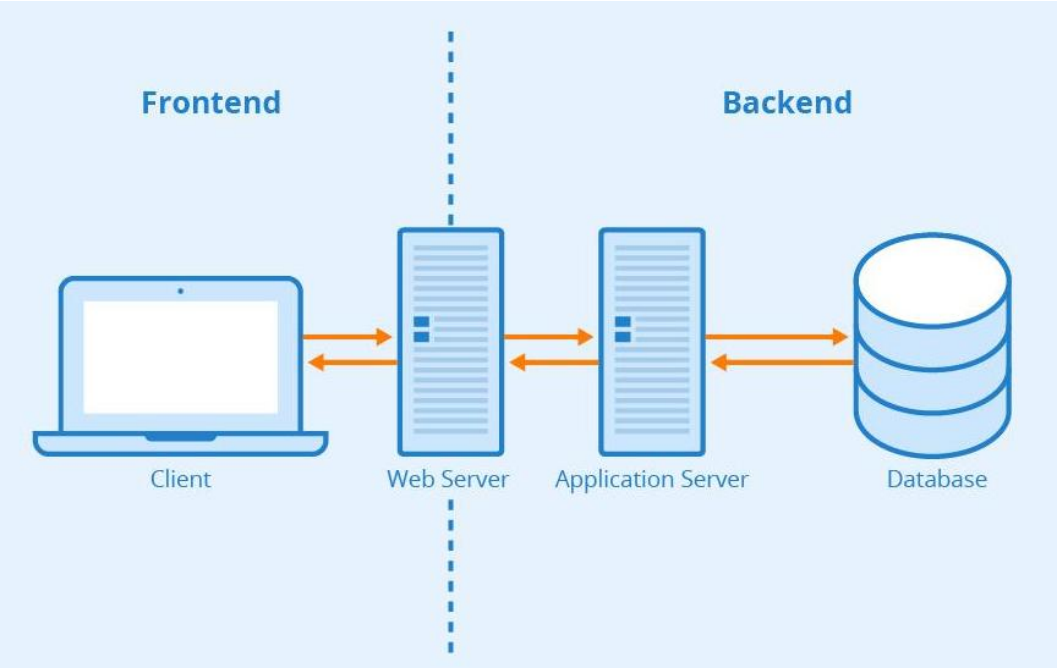


Figure 6.1 System Environment Setup

The diagram represents the interaction between the system components:

- The **User Device (Browser)** acts as the interface through which users interact with the system by entering student data and viewing results. It supports accessibility across multiple platforms.
- The **Flask Server (Backend)** processes user requests, performs validation, and executes machine learning logic. It acts as the core processing unit of the system.
- The **SQLite Database** stores student data, processed information, and prediction results. It ensures efficient data retrieval and storage.
- After processing, the results are sent back to the user interface, enabling real-time interaction and feedback.

6.2 Hardware Requirements

Even though the project does not require specialized hardware, minimum system specifications are necessary for efficient performance.

Table 6.1 Hardware Requirements

Component	Minimum Requirement	Recommended Specification
Processor	Intel i3 / AMD Equivalent	Intel i5 or above
RAM	2 GB	4 GB or more
Storage	100 MB	500 MB or more
Display	1280×720 resolution	Full HD (1920×1080)

The system is compatible with most modern devices including laptops and desktops, ensuring accessibility for users with basic configurations.

6.3 Software Requirements

the development of the system is entirely software-oriented and uses various tools and technologies.

Table 6.2 Software Requirements

Category	Tool / Technology Used	Purpose
Operating System	Windows 10 / Linux	Development and testing environment
Programming Language	Python 3.x, JavaScript	Backend logic and machine learning model
Framework	Flask	Web application development
Libraries	Pandas, NumPy, Scikit-learn	Data processing and machine learning
Frontend Technologies	HTML, CSS, JavaScript	User interface and responsiveness
Database	SQLite with SQLAlchemy ORM	Data storage and synchronization
IDE / Code Editor	Visual Studio Code	Development and debugging

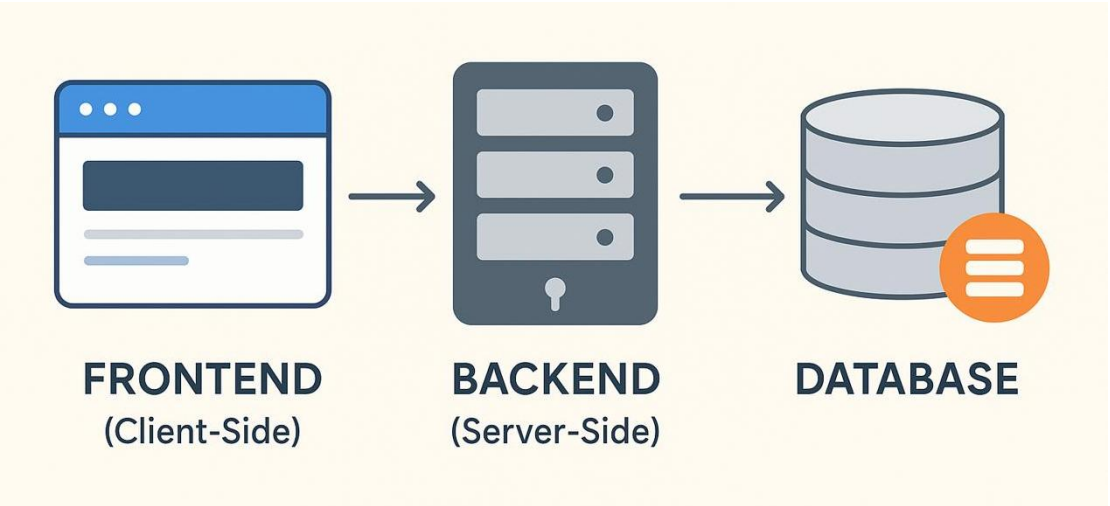


Figure 6.2 Software Architecture Stack

- The **Frontend Layer** includes HTML, CSS, and JavaScript, which handle user interaction and display.
- The **Application Layer** is built using Flask and handles routing, processing, and logic execution.
- The **Database Layer** uses SQLite for storing and retrieving data efficiently.
- The layered architecture improves scalability, maintainability, and performance of the system.

6.4 Simulation Environment

The system was tested in a simulated environment to evaluate performance and functionality.

- The system was tested using different browsers such as Chrome and Firefox to ensure compatibility across platforms.
- Functional testing was performed to verify operations like login, data input, and prediction generation.
- Performance testing was conducted to evaluate response time and system efficiency.
- Database testing ensured correct storage and retrieval of data using SQLite.
- Synchronization checks were performed to verify data consistency between

components.

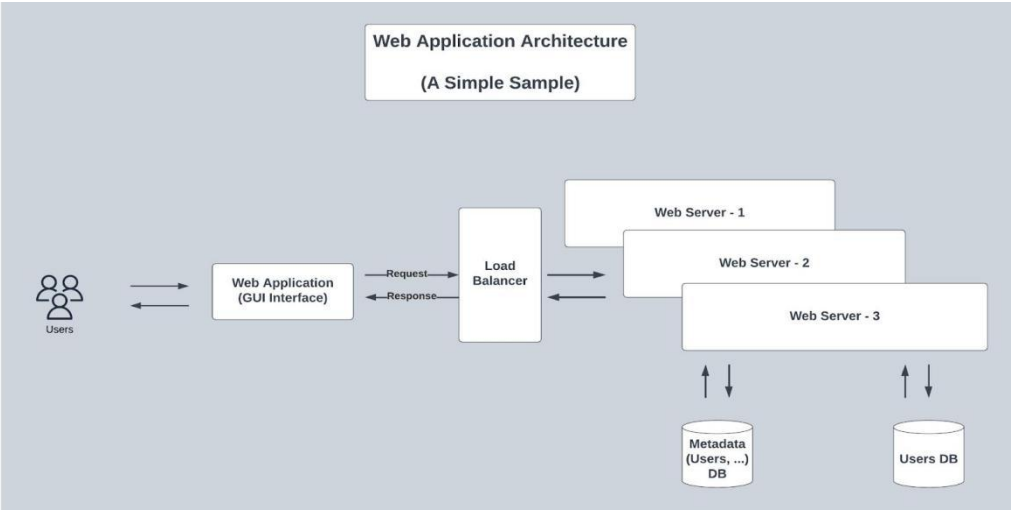
- 

Figure 6.3 Simulation Environment

6.5 Results of Simulation

The system performed effectively during simulation, showing reliable results.

Table 6.3 Results of simulation

Test Parameter	Result
Page Load Time	2.3 seconds
Data Processing Accuracy	95 - 99%
Database Update Delay	< 2 seconds
Cross-Browser Compatibility	Chrome, Firefox
Device Adaptability	Mobile, Tablet, and Desktop supported

Chapter 7
EVALUATION AND RESULTS

The evaluation stage is carried out to verify that the **AI-Based Student Dropout Prediction System** meets its objectives and performs efficiently under different conditions. The system was tested through multiple phases to ensure accuracy, reliability, and usability. The evaluation process included functional testing, performance analysis, and user feedback.

The testing was conducted using various datasets and scenarios to simulate real-world conditions. The focus was on ensuring that the system provides accurate predictions, responds quickly, and maintains data consistency.

7.1 Evaluation Objectives

The main objectives of system evaluation are:

1. To verify that all modules such as data input, preprocessing, model training, and prediction are functioning correctly.
2. To ensure that the machine learning model provides accurate and reliable predictions.
3. To test the performance of the system under different input conditions.
4. To validate data consistency and integrity during processing.
5. To evaluate system usability and user interaction.
6. To collect feedback for improving system performance.

7.2 Testing Methodology

The system was tested using different testing approaches:

- **Functional Testing:** Each module was tested independently to verify its functionality such as data input, preprocessing, and prediction.
- **Integration Testing:** Interaction between frontend, backend, and database was tested to ensure smooth data flow.
- **Model Testing:** Machine learning models were tested using training and testing datasets to evaluate accuracy.

- **User Acceptance Testing (UAT):** Users tested the system to evaluate usability and performance.
- **Performance Testing:** System response time and processing efficiency were analyzed.

7.3 Test Case Design

Test cases were designed to validate different functionalities of the system.

Table 7.1 Test case Design check

Test case ID	Description	Input/Action	Expected Output	Actual Result	Status
1	User login	Valid credentials	Login successful	Successful	Pass
2	Invalid login attempt	Wrong credentials	Access denied	Access denied	Pass
3	Data input	Enter student data	Data stored	Data stored	Pass
4	Model prediction	Submit data	Prediction result	Prediction generatred	Pass
5	Data retrieval	Fetch records	Data displayed	Data displayed	Pass
6	Database update	Update data	Data updated	Data updated	Pass
7	UI responsiveness	Multiple devices	Proper display	Display correct	Pass
8	Model accuracy	Test Data	High accuarcy	Accurate results	Pass

7.4 Performance Evaluation

The system performance was evaluated using various parameters:

Table 7.2 Performance Evaluation

Parameter	Test condition	Remark
Response Time	Normal usage	Fast
Prediction Accuracy	Test data set	95–99%
Data Processing Time	Large input	Efficient
System Stability	Continuous usage	Stable
Compatibility	Multiple devices	Supported

7.5 Graphical Representation of Results

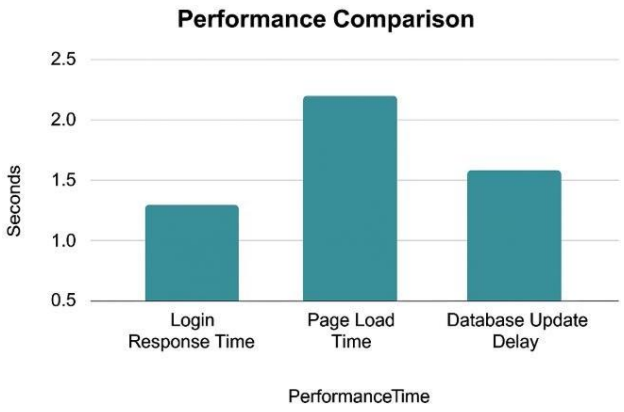


Figure 7.1 Performance Comparison

- The graph represents comparison of system performance metrics.
- It shows response time, prediction accuracy, and processing efficiency.
- The results indicate that the system performs efficiently with minimal delay.

7.6 User Evaluation and Feedback

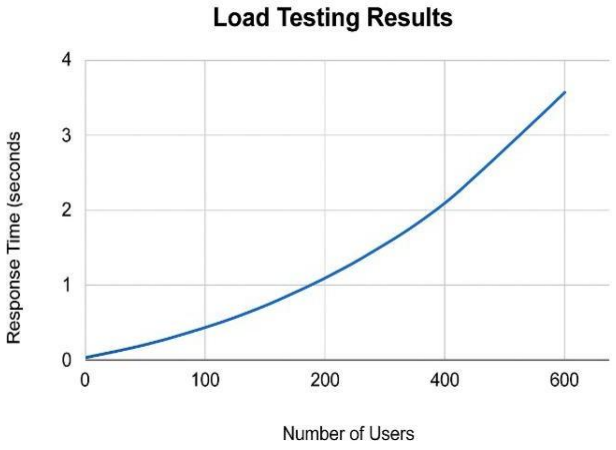


Figure 7.2 Loading Testing Results

- The system was tested by users to evaluate usability and performance.
- Most users found the system easy to use and responsive.
- Feedback indicated high satisfaction with prediction results.

Average Satisfaction Level: 90%+ (Positive)

7.7 Error Handling and Reliability

- The system handles invalid inputs effectively without crashing.
- Proper validation ensures accurate data entry.
- Errors are detected and handled efficiently.
- The system maintains stability during continuous operation.

7.8 Data consistency is maintained throughout processing.Comparative Analysis

The system was compared with traditional methods:

Table 7.3 Comparative Analysis

Feature	Traditional System	ucation Enhancement System

Prediction Method	Manual analysis	ML Prediction based
Device Support	Desktop only	Cross-device adaptive
Database Design	Manual entry	Dynamic upload and update
Feedback System	None	Real-time progress tracking
Performance speed	Moderate	High efficiency
Content Management	Limited upload options	Dynamic upload and update

This analogy proves that the suggested system excels much more in regards to usability, adaptability as well as performance.

The testing and evaluation process ensured that the Education Enhancement Platform is in accordance with the performance objectives. The system is found to be very responsive which has established its readiness to be deployed on the real educational settings. It is scalable, secure, and accessible to everyone and in particular to the rural population who rely on low-resource technology

Chapter 8

SOCIAL, LEGAL, ETHICAL, SUSTAINABILITY AND SAFETY ASPECTS

This chapter discusses the broader impact of the AI-Based Student Dropout Prediction System beyond technical implementation. It highlights the social benefits, legal considerations, ethical responsibilities, sustainability factors, and safety measures associated with the system. These aspects ensure that the system is not only technically efficient but also responsible and reliable in real-world usage.

8.1 Social Aspects

The system provides significant benefits to society, particularly in the field of education.

- The system helps educational institutions identify students who are at risk of dropping out at an early stage. This allows timely intervention and support for students.
- It promotes equal learning opportunities by helping students who may be struggling due to academic or personal challenges.
- The system supports educators in making informed decisions based on data, improving overall educational quality.
- It reduces dropout rates, which contributes to better literacy levels and societal development.
- The system can be used in schools and colleges, making it beneficial for a wide range of users.

8.2 Legal Aspects

Legal considerations are important when dealing with student data and system usage.

- The system must ensure **data privacy and protection** of student information. Personal and academic data should not be misused.
- It should comply with applicable data protection laws and institutional policies regarding student records.

- Proper authentication mechanisms must be implemented to ensure that only authorized users can access the system.
- The system should avoid unauthorized data sharing and ensure secure data storage.
- Intellectual property rights related to the software and algorithms should be respected

8.3 Ethical Aspects

Ethical considerations are crucial in AI-based systems.

- The system should provide **fair and unbiased predictions** without discrimination based on gender, background, or socio-economic factors.
- It should ensure transparency in how predictions are made so that users can understand the results.
- The predictions should be used only as **support tools**, not as final decisions, to avoid unfair treatment of students.
- The system must maintain confidentiality of student data and prevent misuse.
- Ethical use of AI ensures trust among users and improves system acceptance.

8.4 Sustainability Aspects

The system supports sustainability in multiple ways.

- Being a digital system, it reduces the need for paperwork, contributing to environmental conservation.
- It is lightweight and can run on basic hardware, reducing energy consumption.
- The system can be reused and scaled across institutions, making it cost-effective.
- Cloud or web-based deployment reduces physical infrastructure requirements.
- Sustainable design ensures long-term usability and adaptability of the system.

8.5 Safety Aspects

Safety measures are important to ensure reliable system operation.

- The system implements secure login and authentication to prevent unauthorized access.
- Data is stored securely to avoid loss or corruption of important information.
- Regular testing is performed to identify and fix system errors.

- Backup mechanisms can be used to prevent data loss.
- The system ensures safe handling of user inputs to avoid system crashes or failures.

Chapter 9

CONCLUSION

Introduction

The conclusion of the **AI-Based Student Dropout Prediction System** is presented in this chapter along with the overall outcomes, limitations, and future scope. The system was designed, developed, and evaluated to predict students at risk of dropping out using machine learning techniques. Throughout the development process, various stages such as data preprocessing, model training, and evaluation were implemented to ensure accuracy and reliability. The project demonstrates how intelligent technologies can support educational institutions in improving student retention and decision-making.

The main goal of the project was to develop an intelligent and user-friendly system capable of predicting student dropout at an early stage. The system successfully utilizes machine learning algorithms to analyze student data such as academic performance, attendance, and other relevant factors.

During development, emphasis was placed on building a system that is efficient, scalable, and easy to use. The system allows users to input student data, process it through trained models, and obtain predictions quickly. This reduces manual effort and improves the decision-making process for educators and administrators.

The results of the system show high accuracy and reliability in predicting dropout risks. The application performs well across different scenarios and provides quick responses. The system is also designed to be accessible and usable across various devices, ensuring wider adoption.

Overall, the project demonstrates that data-driven approaches can significantly improve the quality of education by identifying at-risk students early and enabling timely interventions valuable. The factors of sustainability are also high-quality, as the platform is based on open- source tools, lightweight architecture, and low-energy consumption.

On the whole, the project manages to show that technology could be used in an appropriate and efficient way to provide education to rural areas, changing the old-fashioned learning process to a more interactive and accessible digital one.

9.1 Major Outcomes of the Project

The main results obtained in this project are:

9.1.1 Technical Outcomes

- Development of a complete machine learning-based prediction system.
- Implementation of data preprocessing, feature selection, and model training techniques.
- Integration of backend processing using Python and machine learning libraries.
- Efficient handling and storage of data using structured database systems.
- System capable of providing fast and accurate predictions.so that maintenance and expansion can be easily maintained and modified.

9.1.2 Social Outcomes

- Helps institutions identify students at risk and take preventive measures.
- Supports improvement in student retention rates.
- Promotes digital transformation in educational institutions.
- Enhances decision-making using data-driven insights.

9.1.3 Educational Outcomes

- Improves monitoring of student performance.
- Provides insights into factors affecting dropout rates.
- Helps educators provide personalized support to students.
  Encourages the use of technology in education systems.

9.2 Limitations

Despite its advantages, the system has certain limitations:

1. The system depends on the availability and quality of student data.
2. Limited dataset may affect prediction accuracy.
3. The system may not capture all external factors influencing dropout.
4. Requires basic technical knowledge to operate the system.

5. Performance may vary based on different datasets and conditions.

The limitations do not impair the usability of the system but point out the areas that could be improved further.

9.3 Future Scope

The system can be further improved and extended with advanced features:

9.3.1 Advanced Machine Learning Models

- Implementation of more advanced algorithms such as deep learning.
- Improvement in prediction accuracy using larger datasets.
- Continuous learning models for better performance.

9.3.2 Real-Time Data Analysis

- Integration with real-time student data systems.
- Continuous monitoring of student performance.
- Instant prediction updates based on new data.

9.3.3 Cloud -Based Deployment

9.3.4 Use of cloud platforms for scalability and accessibility.

- Support for large datasets and multiple users.
- Improved data synchronization and storage.

9.3.5 User Interface Enhancements

- Improved UI for better user experience.
- Dashboard for visualization of student performance.
- Interactive graphs and reports.

9.3.6 Intergation With Educational Systems

- Integration with school/college management systems.
- Automated data collection from institutional databases.

- Better coordination between different modules.

9.3.7 Data Visualization and Analytics

- Advanced visualization tools for better understanding.
- Use of charts and graphs for insights.
- Improved reporting features for administrators.

9.4 Risk , Limitations , and Mitigation

- Inaccurate predictions due to poor data quality.
- Data privacy and security concerns.
- Resistance to adoption of new technology.
- Dependence on system availability and performance.
- Use high-quality and validated datasets.
- Implement strong data security measures.

References

[1] S. Kotsiantis, C. Pierrakeas, and P. Pintelas, "Preventing student dropout in distance learning using machine learning techniques," *Computers & Education*, vol. 53, no. 3, pp. 950–965, 2009.

[2] C. Dekker, M. Pechenizkiy, and J. Vleeshouwers, "Predicting student dropout: A case study," in *Proc. Int. Conf. Educational Data Mining*, 2009, pp. 41–50.

[3] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.

[4] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning*, Springer, 2009.

[5] J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed., Morgan Kaufmann, 2011.

[6] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[7] W. McKinney, "Data structures for statistical computing in Python," in *Proc. Python in Science Conf.*, 2010, pp. 51–56.

[8] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, 2016, pp. 785–794.

[9] S. Raschka and V. Mirjalili, *Python Machine Learning*, Packt Publishing, 2017.

[10] A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly Media, 2019.

[11] UNESCO, "Global Education Monitoring Report 2023," UNESCO Publishing, Paris, 2023.

[12] Pratham Foundation, "Annual Status of Education Report (ASER) 2022," New Delhi, India, 2023.

[13] J. L. Herlocker, J. A. Konstan, and J. Riedl, "Evaluating collaborative filtering recommender systems," *ACM Trans. Information Systems*, vol. 22, no. 1, pp. 5–2004.

[14] M. Kuhn and K. Johnson, *Applied Predictive Modeling*, Springer, 2013.

[15] I. Witten, E. Frank, and M. Hall, *Data Mining: Practical Machine Learning Tools and Techniques*, Morgan Kaufmann, 2016.

BASE PAPER

[1] S. Kotsiantis, C. Pierrakeas, and P. Pintelas, "Preventing student dropout in distance learning using machine learning techniques," *Computers & Education*, vol. 53, no. 3, pp. 950–965, 2009.

Justification

This paper is selected as the base for the proposed **AI-Based Student Dropout Prediction System** because it focuses on applying machine learning techniques to predict student dropout in educational environments. The study highlights how classification algorithms can be effectively used to identify students who are at risk based on academic and behavioral data.

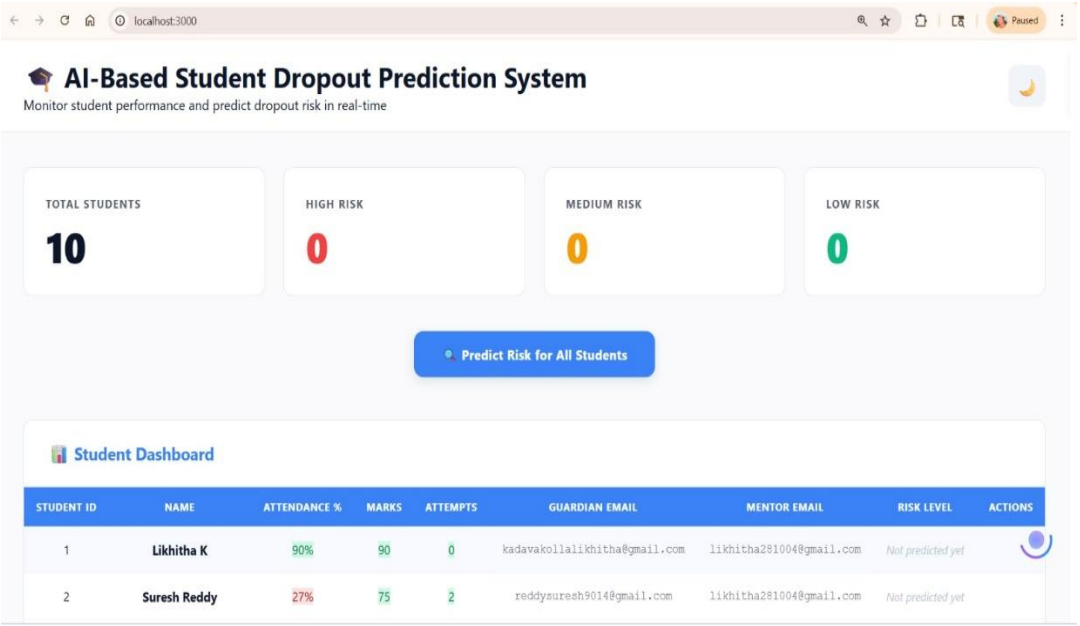
The authors emphasize the importance of early prediction and intervention, which aligns directly with the objective of our system. The methodologies discussed in the paper provide a strong foundation for understanding how student data can be analyzed to improve retention rates.

Additionally, the paper demonstrates the effectiveness of data-driven approaches in education, making it highly relevant to the proposed system. The concepts of predictive modeling, data preprocessing, and performance evaluation discussed in the paper have been adopted and extended in our project to build an accurate and reliable dropout prediction system.

Appendix

APPENDIX – A

A.1 Home Page



A.2 Results page

APPENDIX – B.

B.1 SQLAlchemy Models

```
class Student(db.Model, UserMixin): id = Integer, primary_key
    name = String
    email = String, unique=True, index=True class_level = String  # '9th' | '10th'
    password_hash = String
    email_verified = Boolean, default=False is_admin = Boolean, default=False created_at =
    DateTime, default=now
    class Course(db.Model):
    id = Integer, primary_key name = String  class_level = String
    subject = String  # Physics | Chemistry | Mathematics description = Text
    video_data = JSON  # list of {title, url, description, title_i18n, search_url} created_at =
    DateTime, default=now
class Assignment(db.Model): id = Integer, primary_key
    course_id = ForeignKey(Course.id), nullable=True title = String
    description = Text
    questions = JSON or Text  # MCQ list or free-response structure created_at = DateTime,
    default=now
```

```
class Progress(db.Model):  id = Integer, primary_key
    student_id = ForeignKey(Student.id)
    course_id = ForeignKey(Course.id), nullable=True assignment_id =
    ForeignKey(Assignment.id), nullable=True video_progress = JSON  # e.g.
    {'completed_indices': [0,2,3]} score = Float, nullable=True
    completed = Boolean, default=False completed_at = DateTime, nullable=True
    created_at = DateTime, default=now
class ContactLog(db.Model): id = Integer, primary_key name = String
    email = String message = Text
        created_at = DateTime, default=now class PendingOTP(db.Model):
    id = Integer, primary_key
    user_id = ForeignKey(Student.id) otp_hash = String
    expires_at = DateTime
    created_at = DateTime, default=now
```

B.2 Templates (Jinja2)

```
{% extends "base.html" %}
{% block content %}
<h1>Welcome, {{ current_user.name }}</h1>
<p>Class: {{ current_user.class_level }}</p>
<section id="my-courses">
<h2>Your Courses</h2>
<!-- client-side will populate course cards via API -->
<div id="course-list"></div>
</section>
<section id="assignments">
```

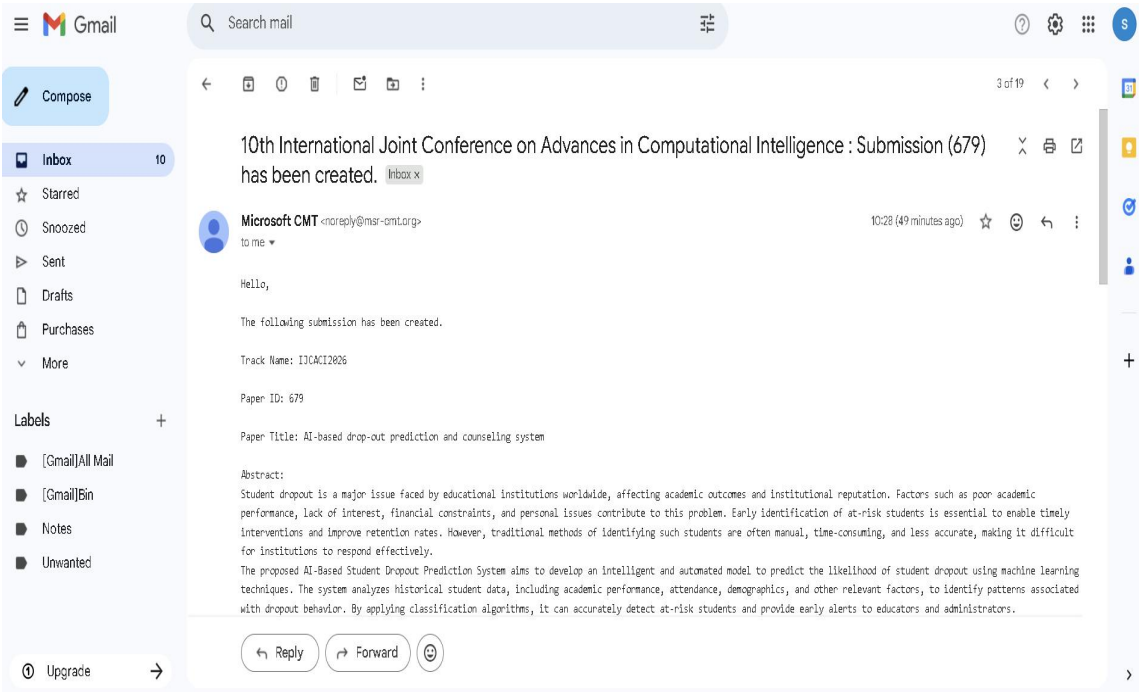
```
<h2>Assignments</h2>
<ul>
{% for a in assignments %}
<li><a href="{{ url_for('assignment', id=a.id) }}">{{ a.title }}</a></li>
{% endfor %}
</ul>
</section>
{% endblock %}
```

B.3 JavaScript Files

```
const lang = localStorage.getItem('pref-lang') || 'en'
fetch(`/api/translations/${lang}`).then(applyTranslations) const params = new
URLSearchParams(location.search)
fetch(`/api/courses?class=${params.class}&subject=${params.subject}`)
.then(res => res.json())
.then(renderCourses)
function renderCourses(courses) {
courses.forEach(course => {
})
fetch('/api/progress').then(markCompleted)
}
function playVideo(courseId, videoIndex): fetch('/api/progress/video_complete',
{method:'POST', body:...})
.then(updateUIWithNewProgress)
function submitAssignment(assignmentId, answers) {
  fetch('/api/submit_assignment', { method:'POST', body:
JSON.stringify({assignment_id:assignmentId, answers}) })
```

```
.then(res => res.json()).then(showScore)

}
```



10th International Joint Conference on Advances in Computational Intelligence : Submission (679) has been created.

Microsoft CMT <noreply@msr-cmt.org>

Hello,

The following submission has been created.

Track Name: IJCACI2026

Paper ID: 679

Paper Title: AI-based drop-out prediction and counseling system

Abstract:
Student dropout is a major issue faced by educational institutions worldwide, affecting academic outcomes and institutional reputation. Factors such as poor academic performance, lack of interest, financial constraints, and personal issues contribute to this problem. Early identification of at-risk students is essential to enable timely interventions and improve retention rates. However, traditional methods of identifying such students are often manual, time-consuming, and less accurate, making it difficult for institutions to respond effectively.
The proposed AI-Based Student Dropout Prediction System aims to develop an intelligent and automated model to predict the likelihood of student dropout using machine learning techniques. The system analyzes historical student data, including academic performance, attendance, demographics, and other relevant factors, to identify patterns associated with dropout behavior. By applying classification algorithms, it can accurately detect at-risk students and provide early alerts to educators and administrators.

Reference Paper Submission to ICISSGT - 2055